Introduction to Proofs

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Introduction to Proofs

Definition

A **proof** is a valid argument that establishes the truth of a mathematical statement. It demonstrates why a statement must be true, not just that it seems true.

- Proofs may use:
  - **Hypotheses** of the theorem (assumptions given),
  - **Axioms** assumed to be true, and
  - **Previously proven theorems**.
- The goal is to apply logical reasoning and rules of inference to reach the desired conclusion.

Purpose

Using these ingredients, a proof validates the truth of the statement being proved through clear and logical steps.

Direct Proof

Direct Proof

A **direct proof** of a conditional statement $p \to q$ assumes p is true and, using axioms, definitions, previously proved facts, and rules of inference, derives q.

Template

1. **Assume** the hypothesis p.
2. **Unpack** relevant *definitions* and known results.
3. **Algebra/Reasoning**: step from the assumptions toward the goal.
4. **Conclude** q.

When to use

Use when the conclusion looks reachable from the hypothesis by straightforward algebra, inequalities, or direct application of definitions.

Definitions: Even, Odd, Parity

Even and Odd

An integer n is *even* if there exist an integer k such that $n = 2k$. It is *odd* if there exist an integer k such that $n = 2k + 1$.

Two integers have the *same parity* if both are even or both are odd; otherwise they have *opposite parity*.

Example 1: Direct Proof

Problem: Prove that if n is an odd integer, then n^2 is odd.

Direct Proof

Assume n is odd. Then there exist an integer k such that $n = 2k + 1$.

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is of the form $2m + 1$ where $m = 2k^2 + 2k$ is an integer. Hence n^2 is odd.

Example 2: Direct Proof

Problem: Prove that if a and b are odd integers, then ab is odd.

Proof

Since a and b are odd, there exist integers r and s such that $a = 2r + 1$ and $b = 2s + 1$. Then

$$ab = (2r + 1)(2s + 1) = 4rs + 2r + 2s + 1 = 2\big(2rs + r + s\big) + 1,$$

which is of the form $2t + 1$ where $t = 2rs + r + s$ is an integer. Therefore ab is odd.

Example 3: Direct Proof

Problem: Prove that every odd integer can be written as a difference of two perfect squares.

Proof

Let an odd integer be $2k+1$ for some integer k. Consider consecutive integers $k+1$ and k:

$$(k+1)^2 - k^2 = (k^2 + 2k + 1) - k^2 = 2k + 1.$$

Hence $2k+1 = (k+1)^2 - k^2$, a difference of two squares. Since every odd integer has the form $2k+1$, the claim follows.

- Example: $11 = 6^2 - 5^2$, $\quad -3 = 1^2 - 2^2$ (works for negative odds too).

Example 4: Direct Proof

Problem: Prove that if m and n are perfect squares, then mn is a perfect square.

Proof

By definition, there exist integers a and b such that $m = a^2$ and $n = b^2$. Then

$$mn = a^2 b^2 = (ab)^2,$$

which is a perfect square (square of the integer ab).

Proof by Contraposition

Proof by Contraposition: Concept

- A **direct proof** starts from the premise p and deduces q to prove $p \to q$.
- Sometimes, a direct approach is difficult or blocked.
- In such cases, we use an **indirect proof**—one common type is the **proof by contraposition**.

Key Idea

The conditional statement $p \to q$ is **logically equivalent** to its contrapositive $\neg q \to \neg p$.

$$p \to q \equiv \neg q \to \neg p$$

Thus, to prove $p \to q$, we can instead assume $\neg q$ and show $\neg p$.

Method

1. Rewrite the statement in the form $p \to q$.
2. Form the contrapositive: $\neg q \to \neg p$.
3. Prove the contrapositive directly.

Example 1: Proof by Contraposition

Problem: Prove that if n is an integer and n^2 is odd, then n is odd.

Contrapositive Form

Original: If n^2 is odd $\Rightarrow$ n is odd.
Contrapositive: If n is even $\Rightarrow$ n^2 is even.

Proof

Assume n is even. Then there exist an integer k such that $n = 2k$.

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2),$$

which is of the form $2m$ for integer $m = 2k^2$. Hence n^2 is even. Therefore, by contraposition, if n^2 is odd, then n is odd.

Example 2: Proof by Contraposition

Problem: Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Contrapositive Form

Original: $3n + 2$ is odd $\Rightarrow$ n is odd.
Contrapositive: n is even $\Rightarrow$ $3n + 2$ is even.

Proof

Assume n is even. Then there exist an integer k such that $n = 2k$.

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1),$$

which is of the form $2m$ where $m = 3k + 1$ is an integer. Hence $3n + 2$ is even.
Therefore, by contraposition, if $3n + 2$ is odd, then n is odd.

Example 3: Proof by Contraposition and Biconditional

Problem: Prove that if n is a positive integer, then n is odd if and only if $5n+6$ is odd.

To Prove

n is odd $\Leftrightarrow$ $5n+6$ is odd.

Proof

We must prove both directions:
(1) If n is odd, then $5n+6$ is odd: Assume $n = 2k+1$. Then

$$5n+6 = 5(2k+1)+6 = 10k+11 = 2(5k+5)+1,$$

which is of the form $2m+1$ for integer $m = 5k+5$. Hence $5n+6$ is odd.
(2) If $5n+6$ is odd, then n is odd: Contrapositive: If n is even, then $5n+6$ is even. Assume $n = 2k$, then $5n+6 = 5(2k)+6 = 10k+6 = 2(5k+3)$, even. Thus both directions hold.

Example 4: Proof by Contraposition

Problem: Show that if n is an integer and $n^3 + 5$ is odd, then n is even.

Contrapositive Form

Original: $n^3 + 5$ is odd $\Rightarrow$ n is even.
Contrapositive: n is odd $\Rightarrow$ $n^3 + 5$ is even.

Proof

Assume n is odd. Then there exist an integer k such that $n = 2k + 1$.

$$n^3 + 5 = (2k + 1)^3 + 5 = 8k^3 + 12k^2 + 6k + 6 = 2(4k^3 + 6k^2 + 3k + 3),$$

which is of the form $2m$ for integer $m = 4k^3 + 6k^2 + 3k + 3$. Hence $n^3 + 5$ is even. Therefore, by contraposition, if $n^3 + 5$ is odd, then n is even.

Proof by Contradiction

Proof by Contradiction: Concept

- In a **proof by contradiction**, we try to prove a statement p by assuming that it is false.
- We then show that this assumption leads to a contradiction, proving that p must be true.

Idea

If we can find a contradiction q such that $\neg p \rightarrow q$ and q is false, then p must be true.

That is, if $\neg p \rightarrow (r \wedge \neg r)$ is true, then p must hold.

Steps

1. Assume the negation of what you want to prove.
2. Use logical reasoning, algebra, or definitions to reach a contradiction.
3. Conclude that the original statement must be true.

Example 1: $\sqrt{2}$ is Irrational (Proof by Contradiction)

Problem: Prove that $\sqrt{2}$ is irrational.

Proof

Assume, to the contrary, that $\sqrt{2}$ is rational. Then there exist integers p and q (with no common factors and $q \neq 0$) such that

$$\sqrt{2} = \frac{p}{q}.$$

Squaring both sides gives $2 = \frac{p^2}{q^2}$, so $p^2 = 2q^2$. This means p^2 is even, hence p is even. Let $p = 2k$ for some integer k. Then

$$(2k)^2 = 2q^2 \Rightarrow 4k^2 = 2q^2 \Rightarrow q^2 = 2k^2.$$

So q is also even. But then both p and q are even, contradicting our assumption that they have no common factor. Therefore, $\sqrt{2}$ is irrational.

Proof by Counterexample

Proof by Counterexample: Concept

- To show that a universal statement $\forall x\, P(x)$ is false, we only need to find one example where $P(x)$ is false.
- Such an example is called a **counterexample**.

Key Idea

To disprove "All x have property $P(x)$," find one x_0 for which $P(x_0)$ is false.

Formally

To show $\neg(\forall x\, P(x))$ is true, show $\exists x\, (\neg P(x))$.

That is, *there exist an integer x such that $P(x)$ is false.*

Example 1: Counterexample

Problem: Show that the statement "Every positive integer is the sum of the squares of two integers" is false.

Proof (by Counterexample)

To disprove the statement, we need to find a positive integer that cannot be written as $a^2 + b^2$ for any integers a and b.
Try small positive integers:

$$1 = 1^2 + 0^2, \quad 2 = 1^2 + 1^2, \quad 3 = ?$$

For 3:

$$a^2 + b^2 = 3 \text{ has no integer solutions since } (0, 1, 2)^2 = \{0, 1, 4\}.$$

Thus, 3 cannot be written as the sum of two squares. Hence, the statement is false.

Proof by Cases

Proof by Cases: Concept

- A **proof by cases** establishes a theorem by dividing it into several mutually exclusive cases.
- Each case must cover all possible situations for the variables involved.
- The statement is proven true by showing that it holds true in every possible case.

Key Idea

To prove a statement P, divide the domain into exhaustive cases $C_1, C_2, \ldots, C_n$ such that:

$$(C_1 \vee C_2 \vee \cdots \vee C_n) \text{ is always true, and } C_i \rightarrow P \text{ for all } i.$$

Then P is true in all cases.

Proof by Cases: Concept

Steps

1. Identify all possible cases for the variable(s).
2. Prove that the statement holds in each case.
3. Conclude that the statement holds for all values by combining the results.

Example 1: Proof by Cases

Problem: Prove that if n is an integer, then $n^2 \geq n$.

Proof

We consider three cases:

Case (i): $n = 0$

$$n^2 = 0^2 = 0 \quad \Rightarrow \quad n^2 \geq n.$$

Case (ii): $n \geq 1$ Multiplying both sides of $n \geq 1$ by n (a positive integer) gives $n^2 \geq n$.

Case (iii): $n \leq -1$ In this case $n^2 \geq 0$, and since $n \leq -1$, we have $n^2 \geq n$.

Because the inequality $n^2 \geq n$ holds in all three cases, we conclude that for every integer n, $n^2 \geq n$.

Example 2: Proof by Cases

Problem: Prove that $|xy| = |x||y|$, where x and y are real numbers.

Proof

Recall that the absolute value of a real number a is defined as:

$$|a| = \begin{cases} a, & \text{if } a \geq 0; \\ -a, & \text{if } a < 0. \end{cases}$$

Because x and y can each be positive or negative, we have four cases to consider.

Example 2: Proof by Cases (continued)

Case (i): $x \geq 0$ and $y \geq 0$ Then $|x| = x$ and $|y| = y$. Hence $|xy| = xy = |x||y|$.
Case (ii): $x \geq 0$ and $y < 0$ Then $|x| = x$ and $|y| = -y$. Because $xy \leq 0$,

$$|xy| = -(xy) = x(-y) = |x||y|.$$

Case (iii): $x < 0$ and $y \geq 0$ Then $|x| = -x$ and $|y| = y$. By symmetry,

$$|xy| = -(xy) = (-x)y = |x||y|.$$

Case (iv): $x < 0$ and $y < 0$ Then $|x| = -x$ and $|y| = -y$.

$$|xy| = xy = (-x)(-y) = |x||y|.$$

Because $|xy| = |x||y|$ holds in all four cases, we conclude that the equality is true for all real numbers x and y.

More Examples

Example 1: More Examples

Problem: Prove that if x is irrational, then $\frac{1}{x}$ is irrational.

Proof by Contradiction

Assume, to the contrary, that $\frac{1}{x}$ is rational. Then there exist integers a and b (with $b \neq 0$ and a, b are co-primes) such that

$$\frac{1}{x} = \frac{a}{b}.$$

Taking the reciprocal of both sides gives

$$x = \frac{b}{a}.$$

Since a and b are integers, x would be rational — a contradiction. Therefore, if x is irrational, $\frac{1}{x}$ must also be irrational.

Example 2: More Examples

Problem: Prove that $m^2 = n^2$ if and only if $m = n$ or $m = -n$.

Biconditional Proof

We will prove both directions.

(1) If $m = n$ or $m = -n$, then $m^2 = n^2$:

If $m = n$, then $m^2 = n^2$.

If $m = -n$, then $m^2 = (-n)^2 = n^2$.

(2) If $m^2 = n^2$, then $m = n$ or $m = -n$: Subtract n^2 from both sides:

$$m^2 - n^2 = 0.$$

Factor using the difference of squares:

$$(m - n)(m + n) = 0.$$

Thus, either $m - n = 0$ (so $m = n$), or $m + n = 0$ (so $m = -n$). Hence, $m^2 = n^2$ if and only if $m = n$ or $m = -n$.

Example 3: More Examples

Problem: Show that the following statements about integer x are equivalent:

(i) $3x + 2$ is even,

(ii) $x + 5$ is odd,

(iii) x^2 is even.

Proof of Logical Equivalence

We show that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii): If $3x + 2$ is even, then $3x$ is even $\Rightarrow$ x is even. Thus, $x + 5 = \text{even} + 5 = \text{odd}$.

(ii) $\Rightarrow$ (iii): If $x + 5$ is odd, then x is even. Hence $x = 2k$ for some integer k, so $x^2 = (2k)^2 = 4k^2 = 2(2k^2)$, which is even.

Example 3: More Examples

Problem: Show that the following statements about integer x are equivalent:

(i) $3x + 2$ is even,

(ii) $x + 5$ is odd,

(iii) x^2 is even.

Proof of Logical Equivalence

We show that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

(iii) $\Rightarrow$ (i): If x^2 is even, then x is even (contrapositive of "if x is odd, x^2 is odd"). Let $x = 2k$. Then $3x + 2 = 6k + 2 = 2(3k + 1)$, even.

Since each implies the next, all three statements are equivalent.